MIDTERM REVIEW
CONNECT THE IDEAS

BOONE BOWLES | NOVEMBER 5, 2026

1 | Firms and value | Remember

- Legal form affects liability, taxation, continuity and access to capital.
- Ownership and control can be separate. Incentives and oversight matter.
- Bigger earnings, more sales or a larger firm do not automatically mean more value.
- Ask what creates durable cash flows, and how stakeholders affect them.

1 | Firms and value | Try it

1. A CEO rejects a valuable long-term investment to protect this year's bonus. What is the main concern?

A. Too little depreciation B. An agency conflict C. A shortage of shareholders

2. Which feature most directly helps a corporation pool capital at scale?

A. Transferable ownership shares B. A guarantee of positive profits C. Every owner manages operations

3. An LLC is taxed as a partnership. Which statement fits our class comparison?

A. Its owners necessarily have unlimited liability B. Its profits must face C-corporation double taxation C. Liability protection and pass-through taxation can coexist

1 | Firms and value | Answers

1 | B The pay rule rewards a personal objective that can conflict with long-term owner value.

2 | A Many investors can hold and transfer claims without running day-to-day operations.

3 | C Legal liability and tax treatment are different dimensions. The question explicitly specifies partnership taxation.

2 | Statements and ratios | Remember

- Balance sheet: a date. Income statement and cash-flow statement: a period.
- Compare like dates, units and definitions; book value and market value differ.
- DuPont separates profitability, asset efficiency and the equity denominator.
- Profit is not cash flow: investment in fixed assets and working capital uses cash.

2 | Statements and ratios | Try it

1. Which statement reports the amount of inventory held on a particular date?

A. Income statement B. Balance sheet C. Statement of cash flows

2. Two firms have equal ROE. What does DuPont tell us?

A. Their profit margins must match B. Their asset turnover must match C. Different combinations of the three drivers can produce the same ROE

3. A firm reports positive net income but spends heavily on equipment and inventory. Its FCF can be:

A. Negative B. Necessarily equal to net income C. Positive by definition

2 | Statements and ratios | Answers

1 | B Inventory is an asset balance at a date. Its purchase and sale affect other statements over time.

2 | C Equal products do not require equal factors. Identify which driver differs before interpreting performance.

3 | A Cash investment can exceed after-tax operating cash generation.

Walmart / Target | Calculate first

1. Using the latest-year figures, which firm has the higher current ratio?

A. Walmart, about 1.27× B. Target, about 0.94× C. Target, about 1.06×

2. Which description fits the latest-year net profit margin and asset turnover?

A. Walmart has the higher margin and lower turnover B. Target has the lower margin and higher turnover C. Walmart has the lower margin and higher turnover

3. Target's operating-income coverage of net interest changed from the prior year to the latest year by:

A. Falling from about 13.54× to 11.50× B. Rising from about 11.50× to 13.54× C. Remaining unchanged

Walmart / Target | Answers 1–3

1 | B Walmart: 84,874 / 107,469 = 0.790×. Target: 20,005 / 21,230 = 0.942×. Neither number alone establishes financial distress.

2 | C Margins: Walmart 21,893 / 713,163 = 3.07%; Target 3,705 / 104,780 = 3.54%. Turnover: 2.51× versus 1.76×, using year-end assets.

3 | A 5,566 / 411 = 13.54×; 5,117 / 445 = 11.50×. Operating income fell while net interest expense rose.

Walmart / Target | Interpret the ratios

4. Which firm had the higher latest-year gross margin using the worksheet's revenue convention?

A. Walmart, about 24.93% B. Target, about 27.93% C. Walmart, about 75.07%

5. Walmart's revenue increased from 680,985 to 713,163. Approximate growth was:

A. 4.73% B. 95.49% C. 3.07%

6. Can these ratios alone establish which stock is the better investment?

A. Yes: choose the higher profit margin B. Yes: choose the higher asset turnover C. No: price, expected future cash flows and risk also matter

Walmart / Target | Answers 4–6

4 | B Walmart: (713,163 – 535,395) / 713,163 = 24.93%. Target: (104,780 – 75,511) / 104,780 = 27.93%. Cost classifications differ across retailers.

5 | A (713,163 – 680,985) / 680,985 = 4.73%. Use the prior-year amount in the denominator.

6 | C Historical ratios describe operations and financial position. Investment value also depends on what you pay and what happens next.

3 | Time, bonds and interest rates | Remember

- Draw dates first. Move every cash flow to the same valuation date.
- Match periods: semiannual coupon, semiannual yield, twice as many periods.
- For fixed promised payments: a higher required yield means a lower price.
- Risk structure compares similar maturities; term structure compares maturities.

3 | Time, bonds and interest rates | Try it

1. A $1,000 bond pays a 5% annual coupon semiannually and has 7 years left. At 6.5% annual YTM, what is its price just after a coupon?

A. $953.71 B. $916.71 C. $917.73

2. You receive $5,000 at each year-end in Years 4, 5 and 6, and nothing else. At 8% annually, what is the value today?

A. $12,885.48 B. $9,471.22 C. $10,228.91

3. Investors view a particular bond as less risky, with other influences fixed. What happens?

A. Demand rises; price rises; yield falls B. Supply rises; price falls; yield rises C. The coupon rate rises automatically

3 | Time, bonds and interest rates | Answers

1 | B N = 14; rate per half-year = 3.25%; PMT = $25; FV = $1,000. Solve PV = –$916.71: the quoted price is $916.71. Use END mode and P/Y = C/Y = 1 for these period inputs.

2 | C First find the Year 3 value: N = 3, I/Y = 8, PMT = 5,000, FV = 0 → $12,885.48. Discount it three more years: 12,885.484936 / 1.08^3 = $10,228.91. Use END mode.

3 | A The demand curve shifts right. The promised coupon stays fixed; price adjusts and yield moves inversely.

Interest rates | Two more checks

1. An upward yield curve with a positive maturity premium proves future short rates will rise. True or false?

A. True B. False C. Only if the bond pays coupons

2. Expected inflation rises, holding other drivers fixed. In the market for nominal bonds:

A. Demand shifts right; supply left B. Both curves shift left C. Demand shifts left; supply right

Interest rates | Answers

1 | B False. A positive term premium can create an upward slope even with flat expected short rates.

2 | C Investors expect lower real returns and borrowers face lower expected real repayment costs. Price falls and nominal yield rises.

One bond | Three prices

- $1,000 face value; 6% annual coupon; semiannual payments; six years remaining.
- 1. Today: annual YTM = 8%. Price the bond.
- 2. Two years later, just after coupon 4: YTM is still 8%. Price again.
- 3. At that same date: YTM falls to 5%. Price again.

Bond price 1 | Today

- N = 12 half-years; PMT = $30; FV = $1,000; yield per period = 4%.

- Price = $906.15. Coupon rate 6% < market yield 8%: a discount bond.

- Calculator: N 12, I/Y 4, PMT 30, FV 1000 → PV −906.15.

Bond price 2 | Two years later

- Four coupons have been paid. Eight remain: N = 8, PMT = $30, FV = $1,000.
- YTM stays 8% annually → 4% per half-year. Price = $932.67.
- At unchanged yield, the discount shrinks as maturity approaches.

Bond price 3 | Same date, lower yield

- N is still 8; PMT is still $30; FV is still $1,000.
- New annual YTM = 5% → 2.5% per half-year. Price = $1,035.85.
- The lower discount rate raises price. Coupon 6% > yield 5%: now a premium.

8.15 | Beryl’s choice

- Keep renting: $50,000 a year, maintenance included.
- Buy current: $150,000 now; $20,000 annual maintenance.
- Buy advanced: $250,000 now; $15,000 annual maintenance; $10,000 annual bottling savings; $35,000 training in Year 1.
- Both: 10-year life; 7-year straight-line depreciation; no salvage; tax 35%; required return 8%.

8.15 | Set it up, then choose

- Use continuing to rent as the baseline. Its incremental NPV is zero.
- Set up Year 0, Year 1, Years 2–7, and Years 8–10 cash flows.
- Pay and deduct training in Year 1, as in our class workbook. Depreciation ends after Year 7.
- Compare NPVs. Does the more advanced machine necessarily create more value?

8.15 | Buy the current machine

- Current: −$150,000 now; $27,000 in Years 1–7; $19,500 in Years 8–10.

- Advanced: −$250,000 now; $19,000 in Year 1; $41,750 in Years 2–7; $29,250 in Years 8–10.

- NPVs: rent $0; current +$19,894; advanced −$9,715.

- Choose the current machine. Larger operating savings do not offset every larger investment.